
Hybrid 8-bit Floating Point (HFP8) Training and Inference for Deep Neural Networks

Xiao Sun

Jungwook Choi*

Chia-Yu Chen

Naigang Wang

Swagath Venkataramani

Vijayalakshmi Srinivasan

Xiaodong Cui

Wei Zhang

Kailash Gopalakrishnan

IBM T. J. Watson Research Center
Yorktown Heights, NY 10598, USA

{xsun, *, cchen, nwang, swagath.venkataramani, viji, cuix, weiz, kailash}@us.ibm.com

Abstract

Reducing the numerical precision of data and computation is extremely effective in accelerating deep learning training workloads. Towards this end, 8-bit floating point representations (FP8) were recently proposed for DNN training. However, its applicability was only demonstrated on a few selected models and significant degradation is observed when popular networks such as MobileNet and Transformer are trained using FP8. This degradation is due to the inherent precision requirement difference in the forward and backward passes of DNN training. Using theoretical insights, we propose a hybrid FP8 (HFP8) format and DNN end-to-end distributed training procedure. We demonstrate, using HFP8, the successful training of deep learning models across a whole spectrum of applications including Image Classification, Object Detection, Language and Speech without accuracy degradation. Finally, we demonstrate that, by using the new 8 bit format, we can directly quantize a pre-trained model down to 8-bits without losing accuracy by simply fine-tuning batch normalization statistics. These novel techniques enable a new generations of 8-bit hardware that are robust for building and deploying neural network models.

1 Introduction

As Deep Neural Networks (DNNs) evolve rapidly and as models get more complex, training times have increased significantly. To mitigate this challenge, efficient training through reduced precision exploitation has become increasingly important. Using reduced precision for data representations and general matrix multiplications (GEMM) can accelerate DNN training dramatically and save significant computing time and power. Indeed, GPUs can already perform mixed-precision training with 16-bit IEEE Half-Precision floating point formats for deep learning tasks [1]. Recently, a new (1-5-2) (sign-exponent-mantissa) floating-point 8-bit format (FP8), was used to successfully train popular ImageNet models [2] without much accuracy loss. In addition, 8-bit Fixed point formats (INT8) have also been explored to train ResNet50 successfully although 1 of the 3 GEMM computations was performed in higher precision [3]. In addition to DNN training, efficient low-precision deployment is critical in a wide range of edge inference use cases where cost and energy constraints can limit

*contributed to this work while at IBM, is currently with the Electrical Engineering Department, Hanyang University, South Korea, email: choij@hanyang.ac.kr